PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

NEXUS

NEXJAM

TRINETRA

Zero-Day Network Threat Detection using Graph Neural Networks & Federated Learning

TEAM

Trinetra

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Problem Statement

TRINETRA IS A REAL-TIME, PRIVACY-PRESERVING CYBER-DEFENSE SYSTEM. INSTEAD OF RELYING ON STATIC "SIGNATURES" OF KNOWN VIRUSES, TRINETRA USES GRAPH NEURAL NETWORKS (GNNS)* TO LEARN THE NORMAL TOPOLOGY OF A NETWORK. IT DETECTS ANOMALIES (LIKE ZERO-DAY ATTACKS OR LATERAL MOVEMENT) BASED ON BEHAVIOR AND USES *FEDERATED LEARNING* TO INSTANTLY SHARE "IMMUNITY" ACROSS DIFFERENT ORGANIZATIONS WITHOUT EVER SHARING PRIVATE DATA.

1.**GAP**: UNKNOWN ATTACKS.* TRADITIONAL FIREWALLS MISS "ZERO-DAY" (NEW) ATTACKS.
MITIGATION: TRINETRA IS UNSUPERVISED; IT FLAGS ANY DEVIATION FROM NORMAL BEHAVIOR, CATCHING ATTACKS THAT HAVE NEVER BEEN SEEN BEFORE.

2.**GAP**: DATA PRIVACY. COMPANIES CANNOT SHARE SECURITY LOGS DUE TO GDPR/PRIVACY LAWS.
MITIGATION:TRINETRA SHARES ONLY MATHEMATICAL MODEL UPDATES, NOT RAW DATA. THIS BREAKS DOWN DATA SILOS WHILE REMAINING LEGALLY COMPLIANT.

ENTERPRISES (BANKS/HOSPITALS):COLLABORATIVE SECURITY WITHOUT RISKING USER DATA PRIVACY.
CRITICAL INFRASTRUCTURE (POWER/WATER):LIGHTWEIGHT PROTECTION FOR IOT DEVICES THAT CANNOT RUN HEAVY ANTIVIRUS SOFTWARE.
SECURITY ANALYSTS:DRASTIC REDUCTION IN "FALSE POSITIVES" DUE TO CONTEXT-AWARE DETECTION.

TECH STACK

SYNTHETIC NETWORK TRAFFIC DATASET (GENERATED IN REAL-TIME)

- NODE-LEVEL FEATURES: REQUEST RATE, CPU USAGE, FAILED LOGINS, DATA TRANSFER VOLUME, CONNECTION COUNT
- OPTIONAL BENCHMARK DATASETS (CICIDS2017 / UNSW-NB15) FOR EXTENDED VALIDATION

DATABASE:

- LIGHTWEIGHT CSV-BASED EVENT LOGGING
- PANDAS DATAFRAME FOR REAL-TIME DATA HANDLING
- (SCALABLE FUTURE SCOPE: ELASTICSEARCH / CLOUD LOG STORAGE)

ALGORITHMS:

- ISOLATION FOREST (UNSUPERVISED ANOMALY DETECTION)
- STATISTICAL THRESHOLD-BASED RISK SCORING
- FEATURE NORMALIZATION & OUTLIER DETECTION

FRAMEWORKS & LIBRARIES:

- PYTHON 3.10+
- SCIKIT-LEARN
- PANDAS & NUMPY
- NETWORKX (NETWORK GRAPH MODELING)
- MATPLOTLIB (VISUALIZATION)
- STREAMLIT (INTERACTIVE DASHBOARD)

SYSTEM MODULES:

- NETWORK TRAFFIC SIMULATOR
- ATTACK INJECTION ENGINE
- REAL-TIME ANOMALY DETECTION ENGINE
- AUTOMATED NODE ISOLATION LOGIC
- ALERT & LOGGING SYSTEM

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WORKFLOW

NOTE: Development Plan During Hackathon (NOT PROJECT WORKFLOW)

STEP 1

Ingest the CICIDS2017 dataset. Preprocess the network flows and convert them into a dynamic graph format (Nodes = IPs, Edges = Flows). Establish the "Normal" baseline.

STEP 2

Build and train the local detection engine using PyTorch Geometric. Implement the Autoencoder to detect anomalies in the graph structure and flow features.

STEP 3

Set up the Federated Learning loop using Flower. Create 3 mock "Clients" and 1 "Server" to demonstrate the weight aggregation and collaborative learning process

STEP 4

Build the Streamlit dashboard to render the live network graph. create a "Red Button" script that injects attack traffic to demonstrate the system detecting and isolating the compromised node in real-time.

ORIGINALITY

HOW NOVEL & CREATIVE IS YOUR SOLUTION AND WHY?

- MOST CURRENT SOLUTIONS LOOK AT PACKETS IN ISOLATION (1D ANALYSIS). TRINETRA IS NOVEL BECAUSE IT ANALYZES THE NETWORK TOPOLOGY (GRAPH STRUCTURE). WE TREAT THE NETWORK AS A LIVING ECOSYSTEM WHERE THE *RELATIONSHIP BETWEEN NODES MATTERS MORE THAN THE INDIVIDUAL PACKET. COMBINING THIS WITH FEDERATED LEARNING MAKES IT A "PRIVACY-FIRST" SYSTEM, WHICH IS A RARE AND CUTTING-EDGE COMBINATION IN CYBERSECURITY.

MARKET COMPETITORS VS. US:

- COMPETITORS: SNORT/SURICATA (RULE-BASED, FAILS AGAINST NEW ATTACKS), DARKTRACE (AI-BASED, BUT CENTRALIZED AND EXTREMELY EXPENSIVE).

OUR DIFFERENTIATOR: TRINETRA IS DECENTRALIZED. WE ALLOW ORGANIZATIONS TO LEARN FROM EACH OTHER. IF A BANK GETS ATTACKED, A HOSPITAL IN OUR NETWORK BECOMES IMMUNE INSTANTLY. WE ARE ALSO OPEN-ARCHITECTURE AND LIGHTWEIGHT ENOUGH FOR IOT.

SCALABILITY

GIVEN 24 HOURS, WHAT OTHER FEATURES WOULD YOU ADD?

- ADVERSARIAL DEFENSE: IMPLEMENT A "GAN" (GENERATIVE ADVERSARIAL NETWORK) MODULE TO PROACTIVELY GENERATE FAKE ATTACKS AND TRAIN THE SYSTEM TO RESIST AI-DRIVEN HACKERS.
- AUTOMATED RESPONSE: INTEGRATE WITH A REAL FIREWALL (LIKE IPTABLES) TO AUTOMATICALLY BAN IPS THE MOMENT THE GRAPH TURNS RED.

HOW SCALABLE IS YOUR PROJECT?

- EXTREMELY SCALABLE. BECAUSE WE USE FEDERATED LEARNING, THE CENTRAL SERVER ONLY RECEIVES TINY MODEL FILES (KILOBYTES), NOT MASSIVE DATA LOGS (TERABYTES). THIS MEANS ONE TRINETRA SERVER CAN SUPPORT MILLIONS OF DEVICES (IOT, EDGE NODES) WITH VERY LOW BANDWIDTH COSTS.

WHY WOULD INVESTORS FUND THIS?

- CYBERCRIME COSTS THE WORLD *TRILLIONS* ANNUALLY. THE BIGGEST BOTTLENECK IS THAT COMPANIES FIGHT ALONE BECAUSE THEY CAN'T SHARE DATA. TRINETRA CREATES A "COLLECTIVE DEFENSE MARKETPLACE" WHERE COMPANIES CAN SUBSCRIBE TO SHARE IMMUNITY WITHOUT SHARING SECRETS. THIS IS THE FUTURE OF ENTERPRISE SECURITY INFRASTRUCTURE.